Cloud Computing

Exercise-4

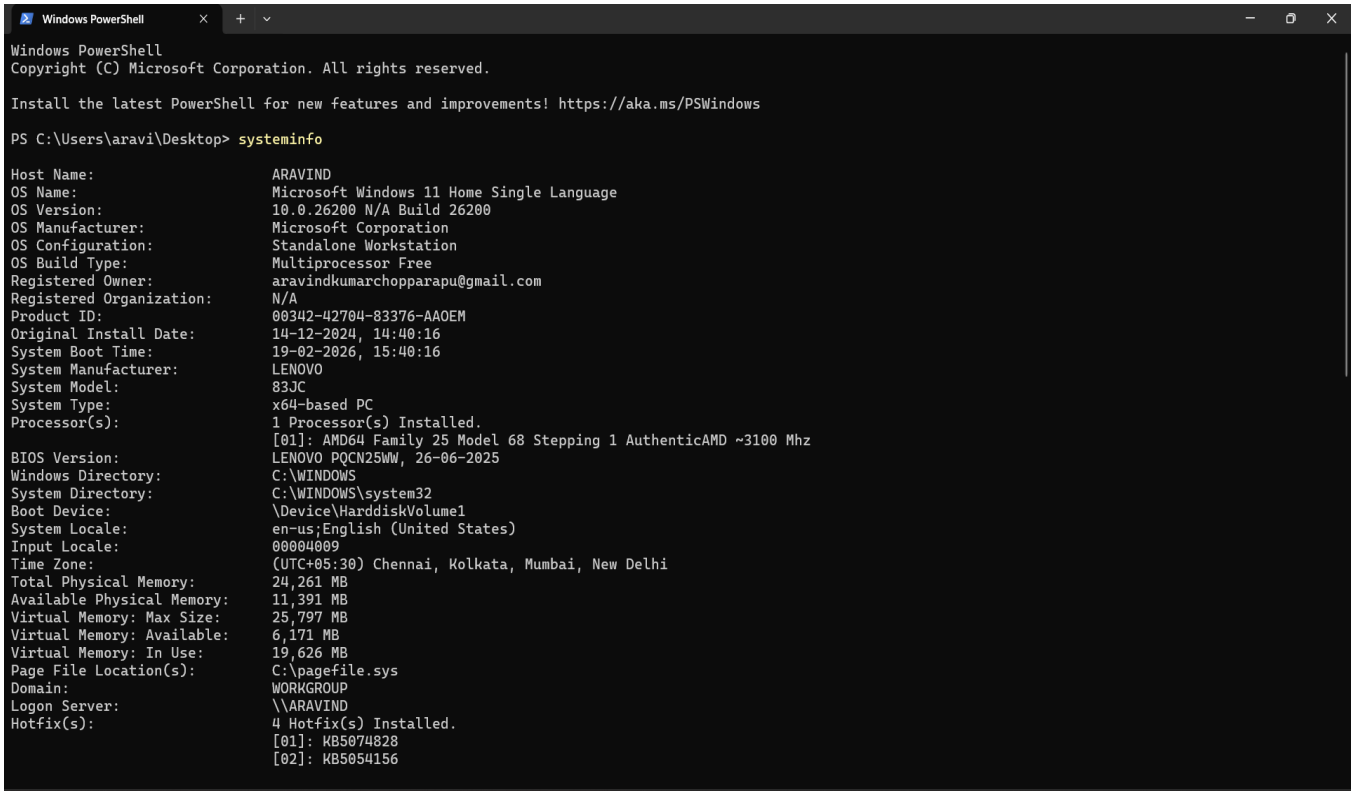
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ID: 23MCCE11

Course: Imtech

1:

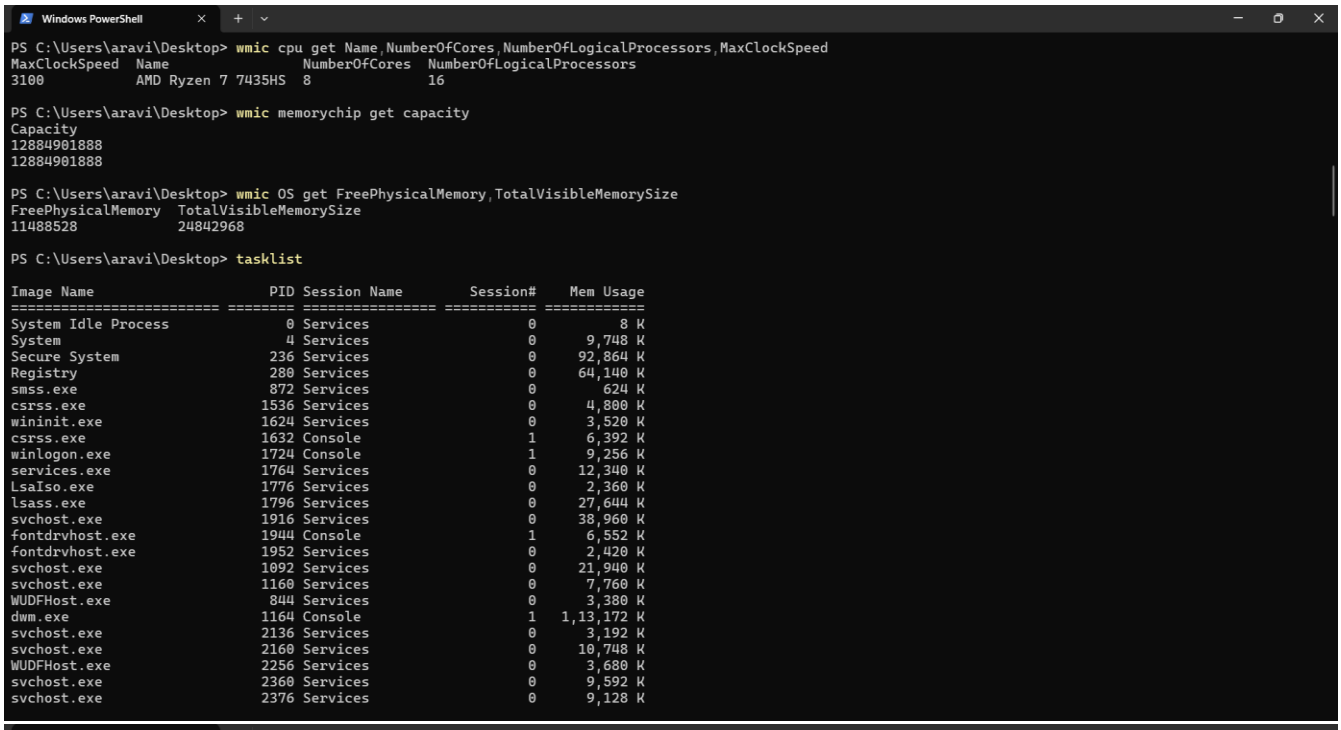
Since there is no mentioning of we must use Linux commands only and I also wanted to try windows commands I am doing this question in windows terminal and to prove that there are done in my laptop is path which has aravi which is short form of Aravind which windows took even I entered my full name as a user.

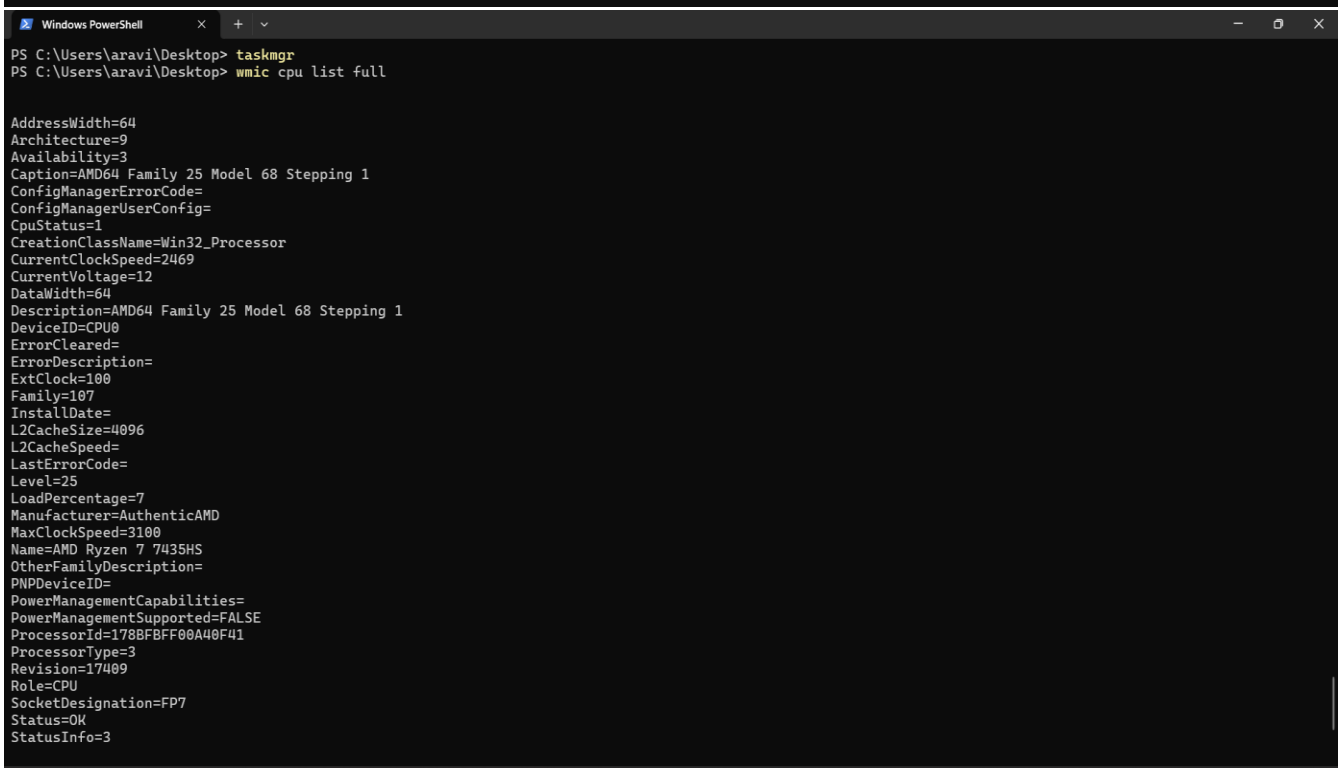


systeminfo gives info about the OS and hardware things. It gives total physical memory and available physical memory also.

Now in Below image we use wmic (windows management instrumentation command line) for 3 commands to specifically get no. of cores, ram, storage

Below those commands tasklist is there which shows list of tasks running like ps aux.

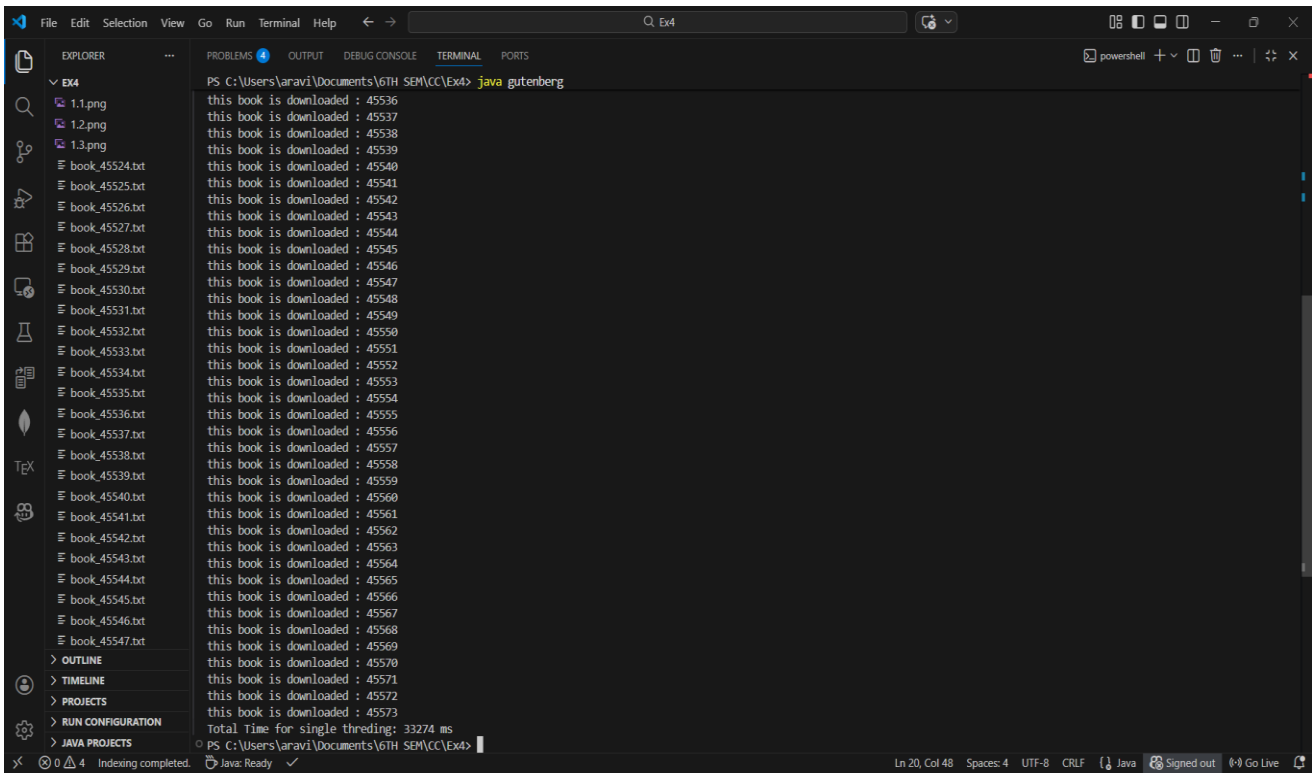




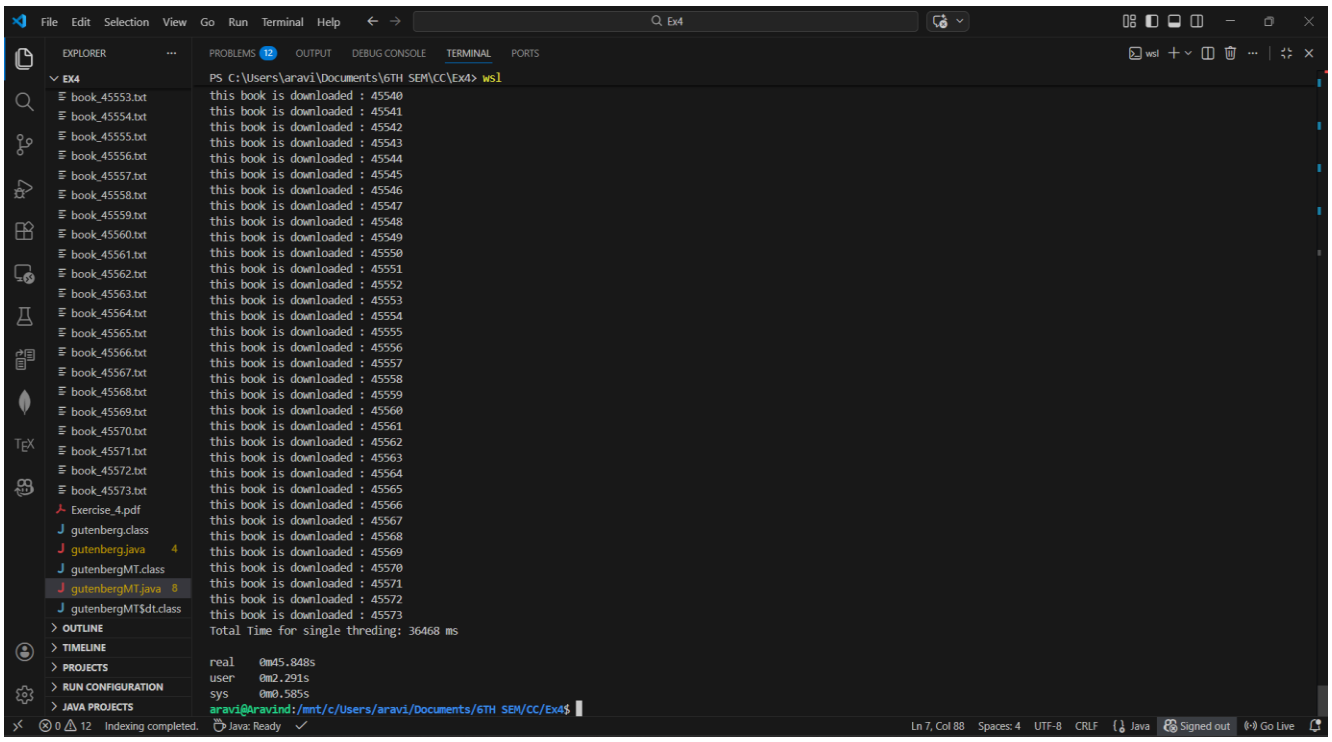
The taskmgr just opens task manager which shows all live information of everything. Below command wmic cpu list full is to get the full info of CPU.

2:

I created 5 java files to for each condition to check which ever file we want



Time taken for single treading 33274 ms = 33.274 S it is compiled in windows terminal itself



It is compiled using wsl for time command

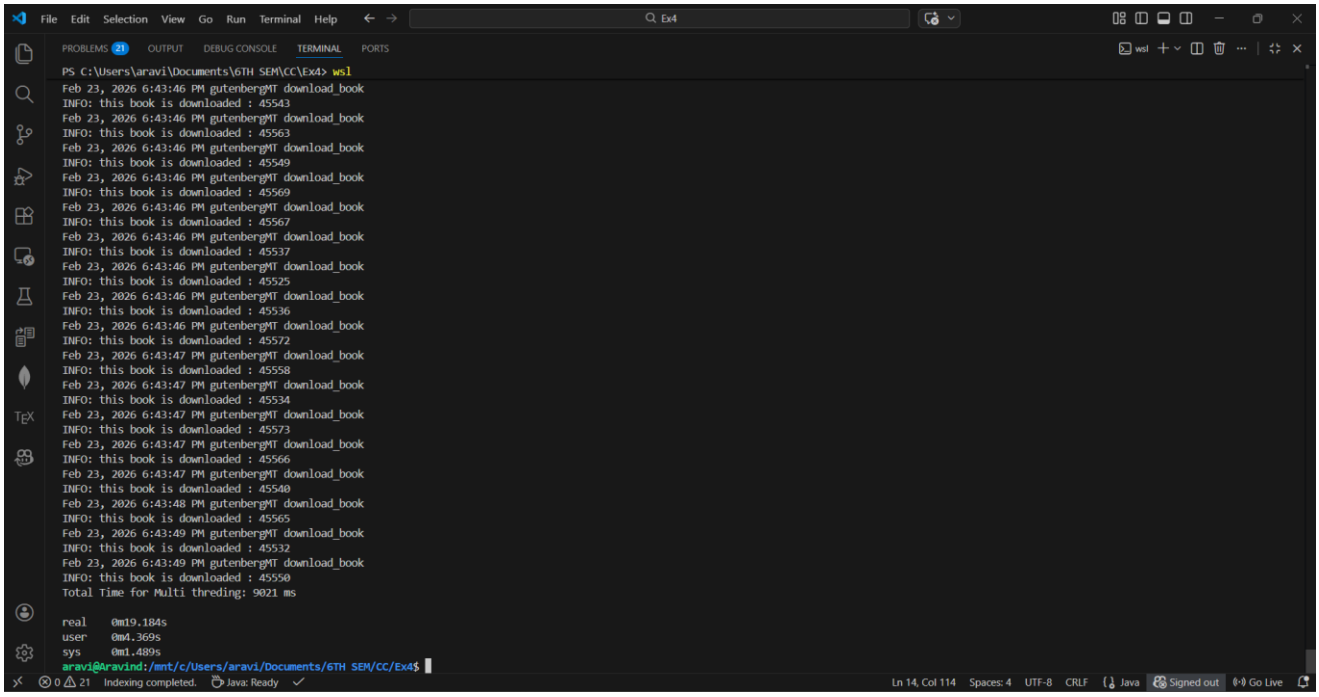
Results:

Time taken to download files: 36468 ms = 36.468 S

Time taken for total code to complete its execution = 45.848 S

Total time in CPU (user) = 2.291 S

Kernal use time = 0.585 S



Results:

Time taken to download files: 9021 ms = 9.021 S

Time taken for total code to complete its execution = 19.184 S

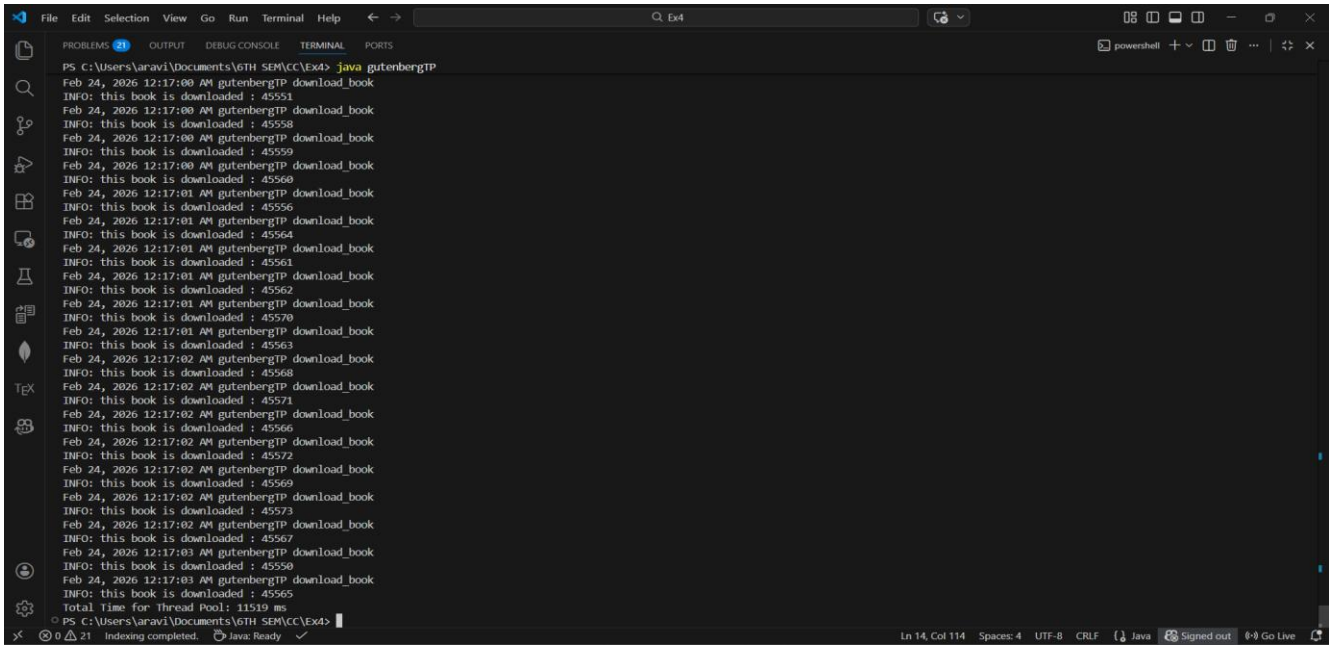
Total time in CPU (user) = 4.369 S

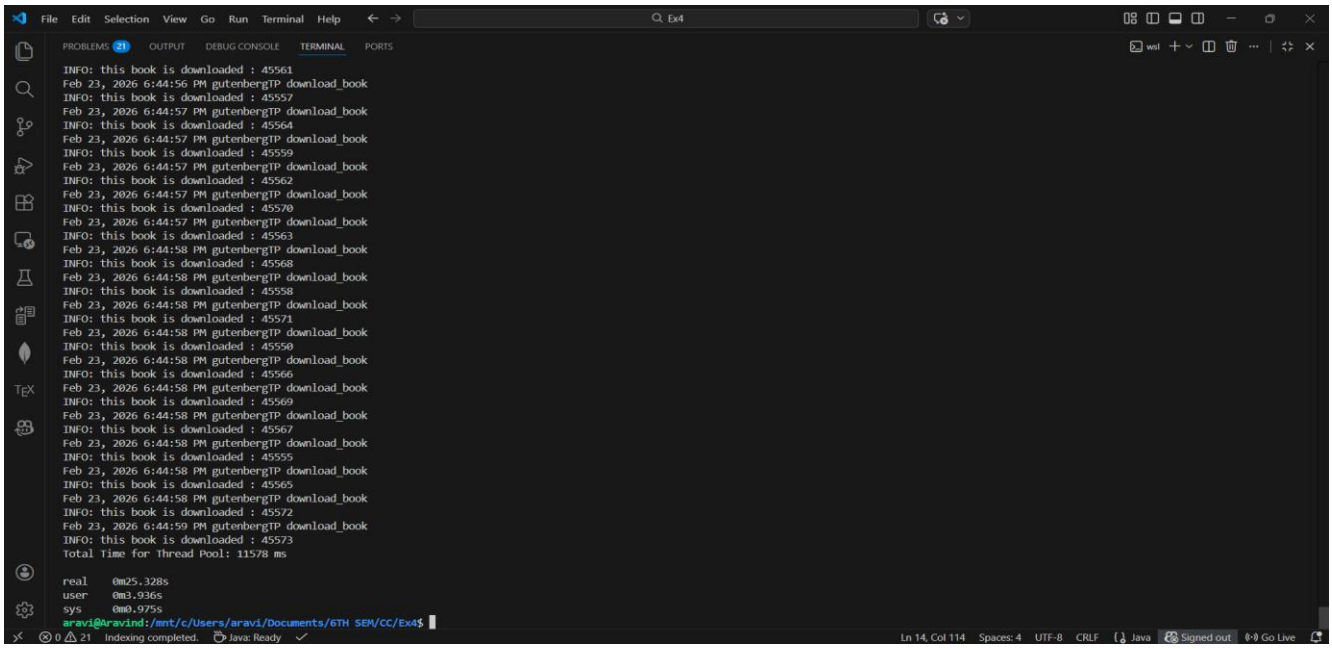
Kernal use time = 1.489 S

It is so much faster compared to single thread which is 36.468 – 9.021 = 27.447 S

In terms of ratio, it is 4.04256734287 times faster according to the results. Results will change from time to time.

By using time commands it took 45 sec for single thread to complete it's execution for multithread it took 19 it is still faster by 45/19 = 2.36 times compared to downloading time it is very much less because of thread initialization.





Results:

Time taken to download files: 11519 ms = 11.519 S

Time taken for total code to complete its execution = 25.328 S

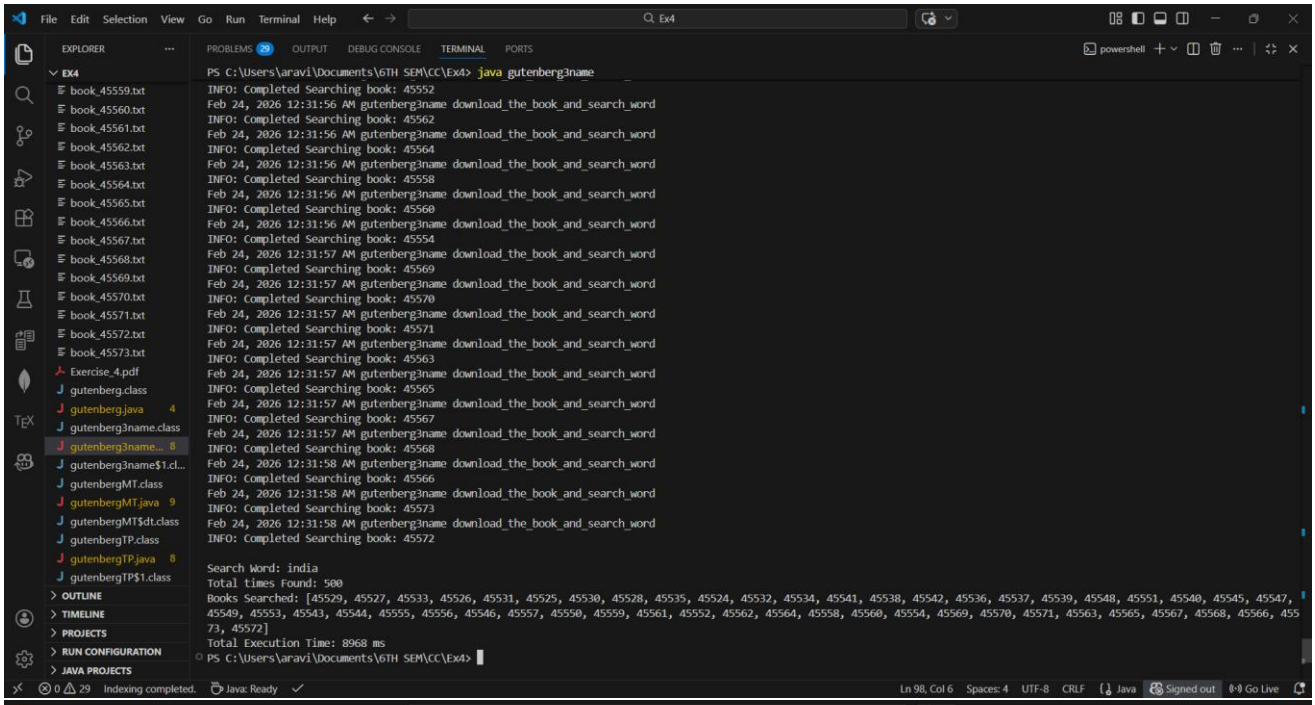
Total time in CPU (user) = 3.936 S

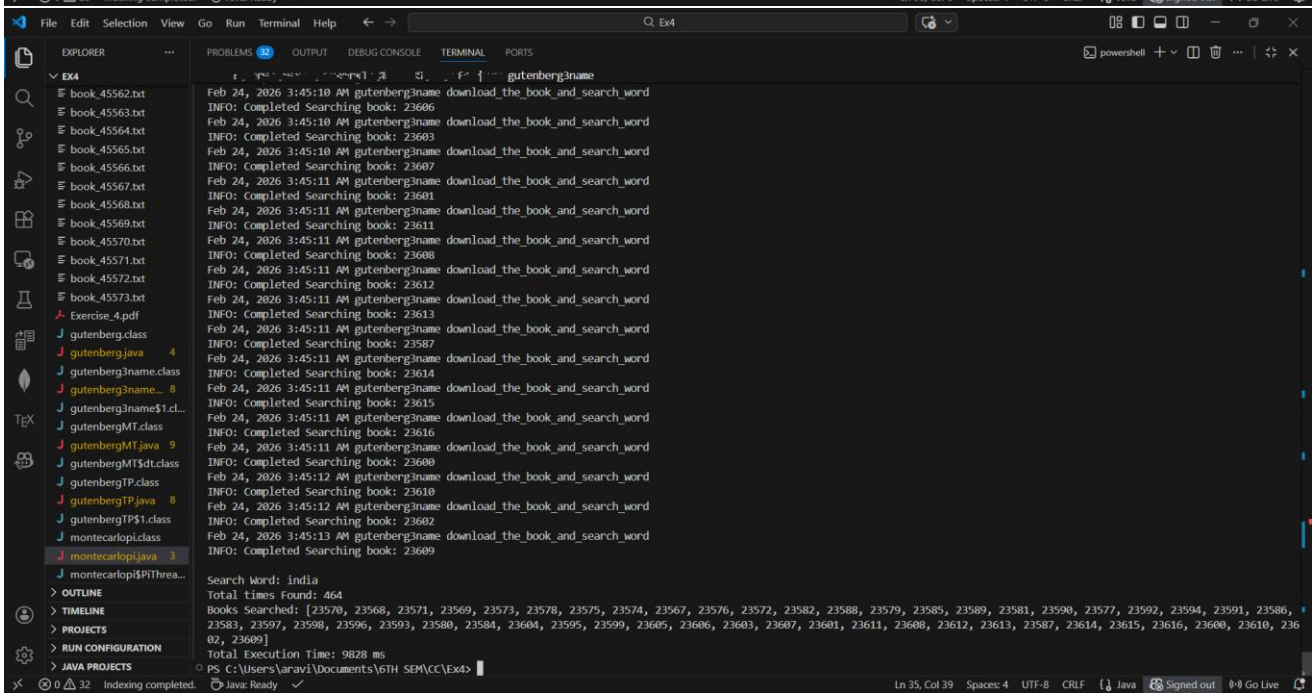
Kernal use time = 0.975 S

Comparing with single thread it is still better in downloading by 3.2 times but in single thread the CPU and kernel use time is lower than both muti threading and thread pool and by comparing with multi-threading it is a bit slower in downloading but less use of Kernal and CPU and code completion time is nearly double to the downloading time except single

threading and is slower by 0.76 times than multi-threading and 1.8 times faster than single thread.

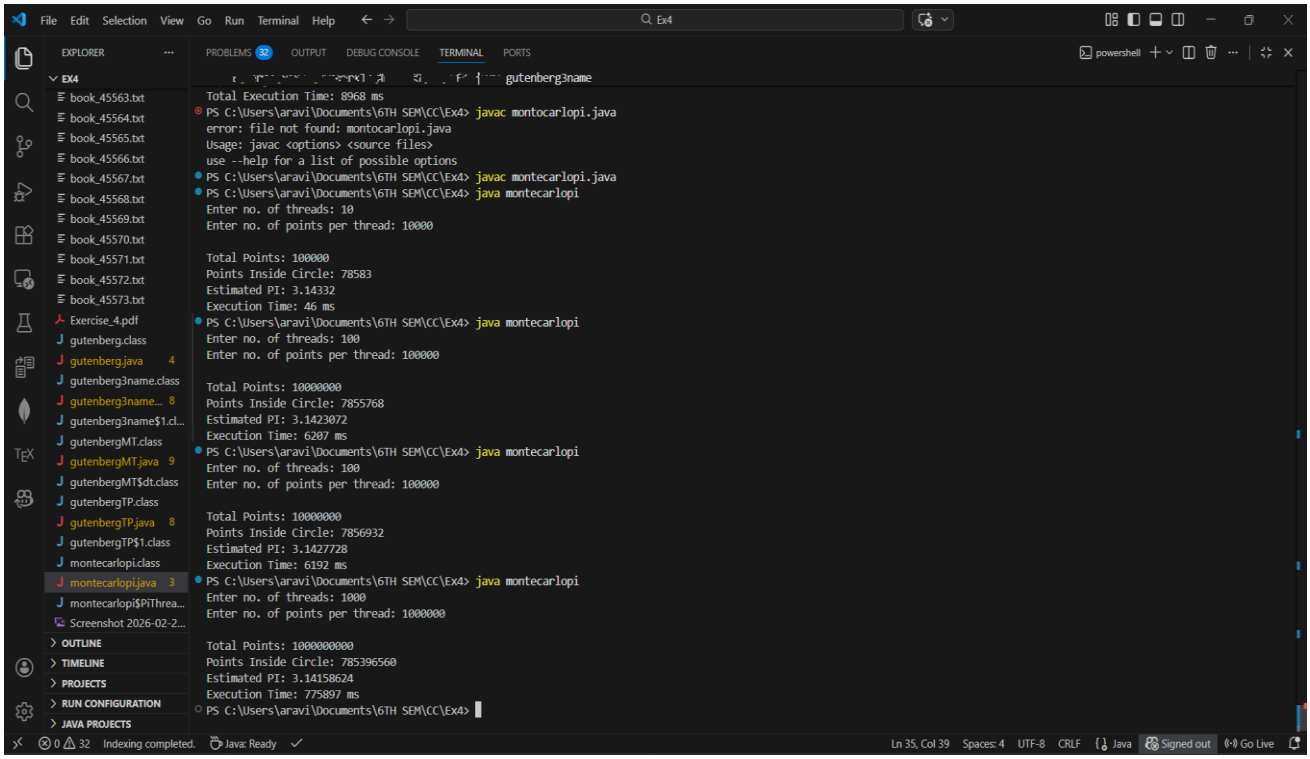
3:





Got the word india count in 2 differnet sets of input one from 45524 to 45573 another from 23567 to 23616 we got their respective results as 500 and 464 . and I checked namsthe for 1st set of files I got zero.

4:



Above are the results of monte carlo pi results. as the no. of points and no. of threads we are getting more accurate values and the time taking to calculate is increased for last calculation it took 775.897 sec which is 12.931 minutes and the value we got is 3.14158624 which is 99.999795849% accurate and the actual pi value is 3.14159265359.